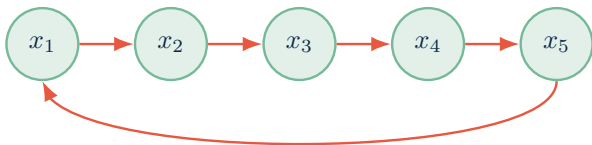


$\sigma = (12345) \in S_5$ permutes the five coordinates



After σ : $x_{\sigma(1)}, \dots, x_{\sigma(5)} = x_2, x_3, x_4, x_5, x_1$

Bring's curve in $\mathbb{P}^4$: $\sum_i x_i = 0, \sum_i x_i^2 = 0, \sum_i x_i^3 = 0$

before σ : $x_1 + x_2 + x_3 + x_4 + x_5 = 0$

after σ : $x_2 + x_3 + x_4 + x_5 + x_1 = 0$

same sum, addition commutes equation unchanged

powers: $\sum_i x_i^2$ and $\sum_i x_i^3$ also symmetric, also fixed by σ

The three defining polynomials are symmetric, hence S_5 -invariant

$\Rightarrow$ Bring's curve maps to itself $\Rightarrow S_5 \subset \text{Aut}(C)$

In fact $\text{Aut}(C) = S_5$ of order 120, the maximum for genus 4 by $|\text{Aut}(C)| \leq 84(g-1) = 252$